Benchmarking solvers for the traveling salesman problem in last-mile delivery

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Objectives

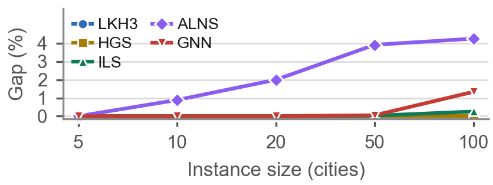
The primary objective of this work is to benchmark five solvers for the Traveling Salesman Problem (ATSP) in the context of a last-mile delivery, comparing their solution quality and runtime. The solvers include three metaheuristics, a hybrid genetic algorithm, and a graph neural network trained by the authors. The secondary objective is to identify the best solver for an application, balancing solution quality and runtime.

Materials and Methods

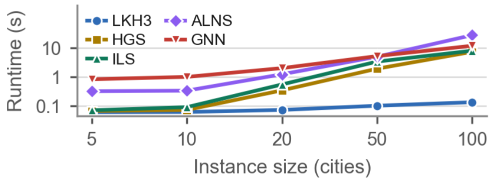
We made a python script which generates random symetric matrices for diferent instance sizes, those matrices were used as the distance matrices for the TSP instances which were then solved by the different algorithms. Five algorithms of distinct families were compared: the Lin–Kernighan–Helsgaun solver (LKH3) of Helsgaun (2017), the Iterated Local Search (ILS) of Lourenço et al. (2003), the Adaptive Large Neighborhood Search (ALNS) of Ropke and Pisinger (2006), the Hybrid Genetic Search (HGS) of Vidal (2022), and a Graph Neural Network (GNN) trained to predict regret values for different edges swap following Kool et al. (2019). The benchmark covers five instance sizes — 5, 10, 20, 50 and 100 cities — with 20 seeds each, all implemented in Python 3.11 and executed on the same machine. Three metrics were recorded per run: the optimality gap, that is, how much the tour cost exceeds the best cost found for that instance by any solver as a percentage; the runtime in seconds; and the win rate: the percentage of runs reaching that best cost.

Results

For up to 20 cities the solvers are essentially indistinguishable in quality, with the single exception of ALNS, which already loses 0.887% at 10 cities and 1.996% at 20, as can be seen in Picture 1. Differences appear only as the instances grow: at 100 cities ALNS reaches a mean gap of 4.251%, the GNN 1.336% and ILS 0.258%, while HGS and LKH3 remain at 0.000%, however, runtime separates them far more sharply, just as seen in Picture 2, where LKH3 scales almost nothing, from about 0.061 s at 5 stops to 0.135 s at 100, at the same time, ALNS climbs to 27.945 s in the last instance. At 100 cities the GNN takes 12.030 s, ILS 8.124 s and HGS 7.432 s.

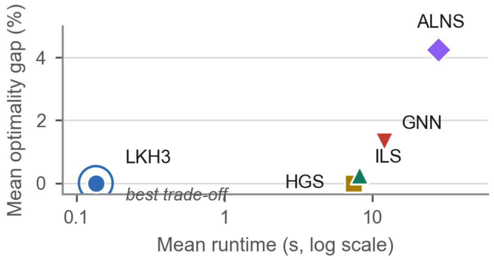


Picture 1: Mean optimality gap of the five solvers across instance sizes.

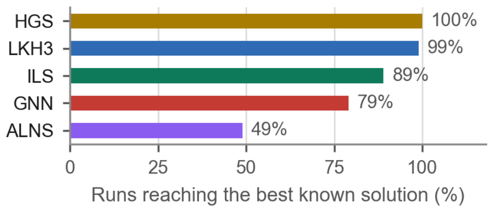


Picture 2: Mean runtime of the five solvers across instance sizes, on a logarithmic scale.

Combining both axes from TSP100's data, it is seen in Picture 3 that LKH3 is the best solver that matches a good mean optimality gap while being more than 50 times faster than HGS, the closest competitor in quality. The win rate shown in Picture 4 evidence that the quality tie between the two is not fully exact since HGS returned the best tour in 100% of the runs against 99% for LKH3, however, the difference is still smaller than 0,005% of the mean gap. It can also be seen that ILS reached the best tour in 89% of the runs, the GNN in 79% and ALNS in only 49%.



Picture 3: Quality versus runtime on the 100-city instances, where LKH3 is the only non-dominated solver.



Picture 4: Percentage of runs in which each solver reached the best known tour, over all instance sizes.

Conclusions

For the chosen set of instance sizes, LKH3 shows up as the best overall solver. It is statistically as accurate as the best alternative while running in a fraction of a second, which is the only behaviour compatible with an interactive application. HGS remains the reference for solution quality and could be the natural fallback whenever runtime is irrelevant. ALNS was the weakest solver on both axes, and the GNN seems to be very promising, especially for larger and asymetrical instances, but not yet competitive at these scales.

The authors declare no conflict of interest.
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